SLIIT
Discover Your Future

BSc (Hons) in Information Technology Specializing Data Science

Research Project - IT4010 Ethical Analysis Report

Group ID: R26-DS-006

Project Title: AI-Based Adaptive VR Meditation System

	Identification and Analysis
Data Privacy Concerns	The system combines four kinds of sensitive data in a VR meditation study (heart signals, visual preferences and session telemetry, voice recordings, and audio preferences), so privacy and responsible data handling are central throughout. The four components are Component B (heart rate / HRV stress inference), Component C (adaptive VR visual environment), Component D (voice-based stress detection), and Component E (adaptive meditation audio). A person's voice and heart rhythm are biometric and can make them identifiable, while stress estimates and preferences can reveal mood and health state; consistent privacy measures are therefore applied across all four components. **Data collected and purpose.** Only data needed for each analysis is used: Component B (heart-rate / HRV stress inference) takes PPG-derived heart-rate variability, skin temperature and a time-of-day value; Component C (adaptive VR visual environment) takes a participant pseudonym, five normalized visual preferences, limited session context and the processed stress estimate; Component D (voice-based stress detection) records roughly 30 seconds of speech before and after a session and derives a stress score; Component E (adaptive meditation audio) takes musical preferences and derived stress summaries and produces meditation audio. No names, emails, raw tracking images or advertising identifiers are used by the adaptive pipeline. **Consent and approval.** The team reports this honestly. Model development used public research datasets under their published academic terms: WESAD and the Hongn et al. wearable dataset for Component B, and MELD for Component D. One small self-collected set of 26 Sinhala voice clips from 7 adult volunteers (Component D) was recorded with their agreement. No large participant study has yet been run; the planned VR pilot of five to seven consenting adults will require informed consent and SLIIT ethics approval before any recording, reconfirmed before each session. The mobile application records a privacy-notice and optional research-consent choice, but an in-app checkbox is not treated as proof of research approval. **Anonymization and pseudonymization.** Personal identity is never attached inside the pipeline. Participants and clips are referenced only by pseudonymous codes (session IDs, participant codes such as P01, or speaker codes such as person1–person7). Any mapping back to a real identity is held separately and restricted; Component C's two-day comparison links the two visits only through a pseudonym, with the identity key kept apart from the telemetry, physiology and analysis files. **Data minimization.** Each component collects the minimum it needs. Component D computes its stress score from the audio itself, so the dialogue transcript is not required for scoring and can be discarded immediately; Component C sends only one resolved five-value preference set and the

	agreed context fields; Component E sends only derived musical instructions, not raw signals. **Component-specific points.** Component B keeps only a rolling buffer of the last 60 heartbeats and holds raw physiology in memory, not on disk. Component C does not receive or store raw PPG, voice, or tracking-camera images, and excludes direct and unique device identifiers from its telemetry. Component D discards raw audio by default (clip logging is off unless a consented validation test is running) and performs speech-to-text locally. Component E sends only a constructed text music prompt to the external model, never raw PPG, account names or heart-rate fields. **Compliance.** Data handling follows SLIIT research ethics guidelines and the Personal Data Protection Act, No. 9 of 2022, of Sri Lanka. In the current prototype, processing is local and session data is kept on the research devices rather than published.
Data Security	**Restricted storage.** Raw datasets, recordings and trained models are kept in gitignored folders and institutional storage, with access limited to group members and supervisors; the shared repository holds only source code. At runtime, Component B (heart-rate / HRV stress inference) keeps an in-memory rolling buffer; Component D (voice-based stress detection) persists only derived session results in a local database and writes raw clips only when logging is explicitly enabled; Component C (adaptive VR visual environment) appends per-session JSON Lines on the Quest device and the relay; and the mobile app and Component E (adaptive meditation audio) use device storage and a Supabase project with owner-scoped row-level security. **Protection of secrets.** Third-party API keys, such as the ElevenLabs text-to-speech key for Component D and the Google Lyria music key for Component E, are kept in server-side environment files that are never committed to version control and never sent to the client. **Secure input handling.** Every backend validates its input before processing: Component B checks schema, finite values and the exact frame size; Component C enforces strict schema, range and message-size checks, duplicate suppression, single-client binding, and a short-lived pairing code with a retrieval token; Component D applies a DSP quality gate to each clip; and model artifacts are fingerprinted with SHA-256 for integrity. **Short retention of temporary data.** Component D discards live audio once a request completes and deletes any logged clips after analysis; Component C treats the Quest and relay files as temporary recovery copies with a proposed deletion cascade once the master record is verified. **Known gaps, stated honestly.** The development transports default to local ws:// and http:// without TLS or API authentication, and there is no application-level encryption at rest on the local databases and JSONL files. These are known weaknesses of a research prototype: real-participant deployment will require authenticated wss:// / HTTPS, role-based access, encrypted server-side storage and regular backups.

	Data egress. Stress and adaptation inference run locally with no cloud analytics. The only data that leaves a device is Component D's scripted companion text (sent to ElevenLabs to be spoken aloud) and Component E's text music prompt (sent to Google's Lyria model), never raw voice, PPG or identifiers. This external processing is governed by the providers' terms and, for the mobile backend, the Supabase data-processing addendum, which must be confirmed before real-user testing.
Bias and Fairness	Fairness is treated as a design goal across all four components, but the team reports measured disparities honestly rather than claiming uniform performance. **Language fairness.** The system is intended for Sri Lankan users, yet the voice-stress model (Component D, voice-based stress detection) is built on a frozen, largely English-trained speech encoder: English is validated and works well (about 92% on real voices), while Sinhala is out-of-distribution and unreliable and Tamil is not yet evaluated. A controlled experiment that added collected Sinhala data reduced held-out Sinhala accuracy (from about 90.9% to 81.8%) and is reported as a negative result. The honest mitigation is the multimodal design: when a voice reading is low in confidence, the system defers to the heart rate estimate from Component B. **Representativeness of training data.** Component B's public datasets are small and narrow (WESAD has 15 mostly young male subjects), and optical PPG on the low-cost sensor is less accurate on darker skin and with motion. Component E's preference catalogue and synthetic users cover only a limited set of musical and cultural tastes. The planned pilot of five to seven adults is too small to support demographic subgroup claims. **Proxy labels and measurement limits.** Several targets are proxies, and this is disclosed: Component B's four intensity levels come from splitting the stress condition by RMSSD tertiles rather than self-report; Component D scores stress primarily from valence because arousal collapses on real, quiet voices; and Component E's signal confidence is not a calibrated probability that the chosen music will relax a person. **Behaviour under uncertainty.** All components fail safely rather than inventing data. Component B reports a merged band (for example 'mild-to-moderate') when unsure and withholds a prediction until a full window is available; Component C rejects stale, duplicate or low-quality windows and holds the current safe environment or proposes no change; Component D rejects poor clips at its quality gate and defers to the heart signal; and Component E suppresses learning and holds its output when confidence or signal quality is low. **Measured honestly, no subgroup claims.** Performance varies substantially between people (Component B's between-subject macro-F1 standard deviation is about 0.13 to 0.16), and no skin-tone, age or sex subgroup analysis has been performed because the datasets are too small; this is stated as a gap, not a result. The team is committed to improving the weaker language in Component D and validating a larger, more diverse sample before any large-scale deployment.

Transparency and Accountability	Each component documents its data, methods and limitations, and the outputs are designed to be checkable with honest statements of uncertainty. **Documentation and reproducibility.** Data, methods, parameters, experiments and limitations are recorded in each component's documentation, the common report and the research papers, with source code and model settings (random seeds, data splits, confidence thresholds) in the shared repository. Negative and unverified results are kept rather than hidden Component D's (voice-based stress detection) ablation study writes up the Sinhala negative result, and unverified figures are marked as such. A single source of truth is imported by the servers and tests (Component D has 148 passing tests; Component B, heart rate / HRV stress inference, uses parity tests), and the telemetry schema (maintained by Component C, the adaptive VR visual environment) is versioned so decisions can be reconstructed. **Explainable, honestly-bounded outputs.** Component B returns the full probability distribution, a confidence value and, when uncertain, a merged band, and tells consumers not to re-derive a label from the argmax, so a 96% confidence is explicitly not a 96% probability that the diagnosis is correct. Component D shows plain-language results with a quality indicator and keeps the raw valence/arousal numbers behind a separate research view. Component E logs proposed and executed actions, predicted reward and bandit uncertainty, and states that signal confidence is not a guarantee that the music will relax the listener. **Known limitations.** The system is research-grade and is not a medical device; no completed user study yet shows that adapting a session to these signals improves outcomes. Component C's visual ranges and bandit settings are engineering-pilot values, not participant-validated optima, and Component E's archived reinforcement-learning simulation results are explicitly not treated as evidence for its current online agent. **Human oversight, roles and fallback.** Each team member is accountable for their own component (Component B for the stress estimate, Component C for the visual policy, Component D for the voice analysis, and Component E for the audio control), with supervisors overseeing the research. The system is decision support that only nudges a meditation session and controls nothing safety-critical; on any failure each component fails safe (no change, a default non-adaptive session, deferral to the heart signal, an audio mute, or a network-independent emergency stop). The use of AI assistance is disclosed as required by the module, consistent with the ACM Code of Ethics.
Impact on Stakeholders	The stakeholders are the participants and end users, the volunteers who contributed data, the research team and supervisors, and, indirectly, the participants behind the public datasets. Clinicians are not intended decision-makers, because the system is a research prototype that does not diagnose or treat any condition. **Participants and users.** The benefit is a more responsive, personalized meditation experience and an objective record of arousal during a session. The risks are that being told or shown one is 'stressed' could itself cause worry, more so for a user who receives an unreliable reading in a weakly supported language, and that the voice check-in's requirement to keep speaking could

	feel coercive. These are mitigated with supportive, plain-language wording, no diagnostic labels, deferral of uncertain readings to the heart signal, and the ability to skip or stop the voice check-in or switch off adaptation without blocking the session. **Physical and psychological safety.** The VR experience is seated, with no artificial locomotion, policy-controlled camera movement, flashing, particle bursts or fast near-face motion; visual changes are gradual and bounded, and a network-independent emergency stop is always available. The artificial-stress task planned for the pilot must be ethics-approved and standardized, with stopping criteria and debriefing, and the audio uses volume limits and an emergency mute. **Autonomy and rights.** Participants can decline research participation without any implied efficacy claim, stop or mute a session at any time, and understand that their preferences and physiological summaries are linked only through a pseudonymous identifier. The outputs must not be used to monitor, profile or evaluate people, for example in employment, education or discipline. **Underserved language communities.** Sinhala and Tamil users must not silently receive worse results than English users. The limitation is disclosed, mitigated by deferring low-confidence voice readings to the heart signal, and improving the weaker languages is a committed piece of future work before any wider use. **Overall ethical position.** The group system is a research prototype with useful technical evidence and unresolved limitations, so responsible use currently means controlled, expert-supervised demonstrations. Before wider adoption the team will complete informed consent and SLIIT ethics approval, security hardening (TLS, authentication and encryption at rest), probability calibration and subgroup validation, and explicit human-oversight procedures, consistent with the ACM Code of Ethics.